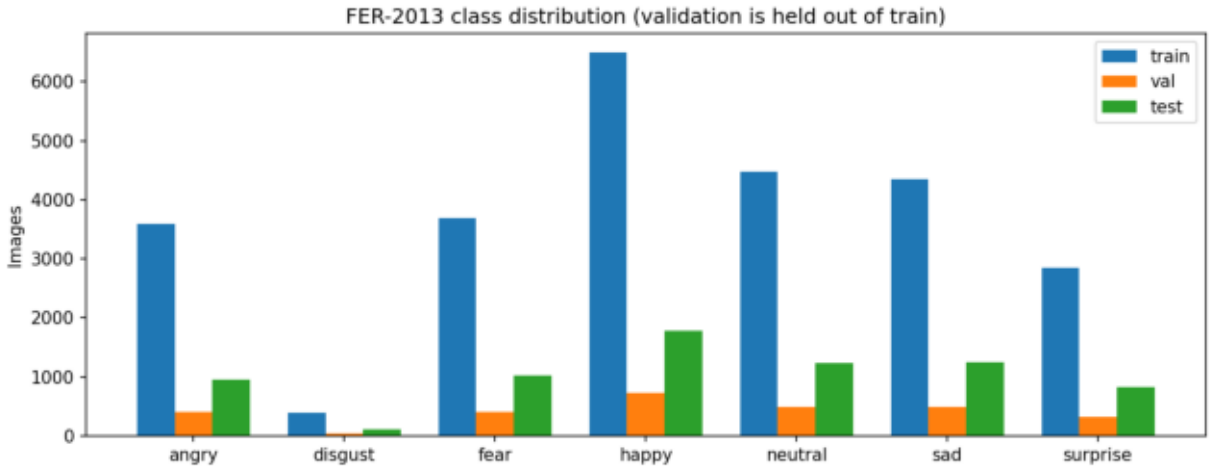
Facial emotion recognition - methods

Data: msambare/fer2013 on Kaggle. Seven emotions; grayscale 48 x 48 pixels. M2 starts from the official DDAMFN++ AffectNet-7 checkpoint; this uses external expression-labeled pretraining. No extra data is mixed into FER fine-tuning.

Split protocol: validation is stratified from the original training folder; the supplied test folder is reserved for final evaluation. The saved split manifests record every file and label. Model selection uses validation accuracy, with macro F1 as a tie-breaker.

Split sizes: train=25,838, val=2,871, test=7,178.
Class order: angry, disgust, fear, happy, neutral, sad, surprise.



Preprocessing M1: grayscale 48x48, scale to [0,1], normalize mean/std 0.5. M2: repeat grayscale to three channels, resize to 112x112; use ImageNet mean/std to match the pretrained model. Upsampling adds no image detail. Validation, test and realtime inference use deterministic preprocessing. Square-root inverse training class frequencies weight the loss. M1 gentle combined: flip, reflect-padded crops, small affine changes and mild jitter. M2 fer_moderate: flip, small affine changes, occasional lighting jitter and small erasures. Selected profiles: M1=gentle_combined, M2=fer_moderate.

M1: four Conv3x3-BatchNorm-ReLU-MaxPool2x2-Dropout blocks, with 32/64/128/256 channels and dropout 0.10/0.15/0.20/0.25. Flatten 2304, Linear256-BatchNorm-ReLU-Dropout0.50, then Linear7.

M2: DDAMFN with mixed-kernel depthwise face features and two coordinate attention heads, then a seven-class linear output. AffectNet classifier rows are remapped to our emotion order. Fine-tune all layers with weighted cross entropy; this is not a reproduction of the upstream SAM/attention-diversity training recipe.

Both models return logits for cross entropy; softmax is applied at inference.

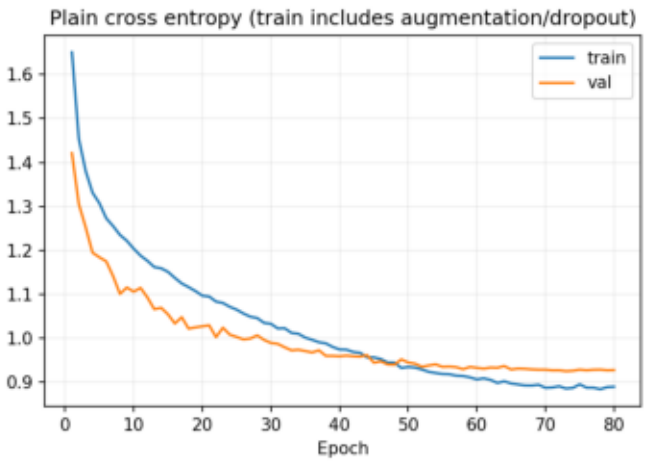
Training and quantitative comparison

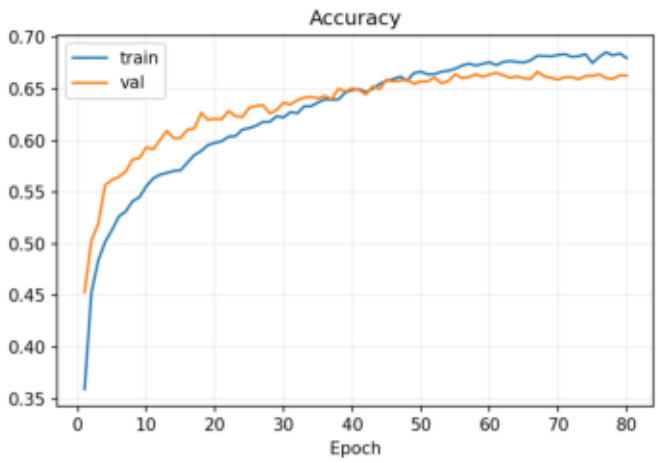
CNN: AdamW lr=0.001, backbone lr=n/a, weight decay=0.0001, smoothing=0.05, epochs <= 80, patience=18, seed=42. Cosine LR schedule. Best epoch=67. Training device=cuda. Augmentation=gentle_combined.

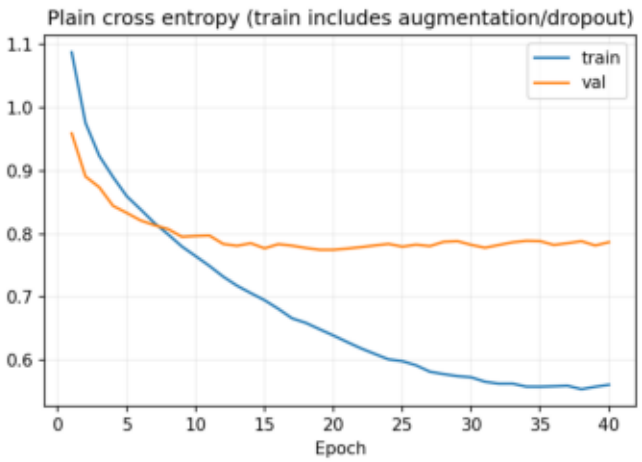
TRANSFER: AdamW lr=0.0001, backbone lr=1e-05, weight decay=0.0001, smoothing=0.02, epochs <= 40, patience=10, seed=42. Cosine LR schedule. Best epoch=32. Training device=cuda. Augmentation=fer_moderate.

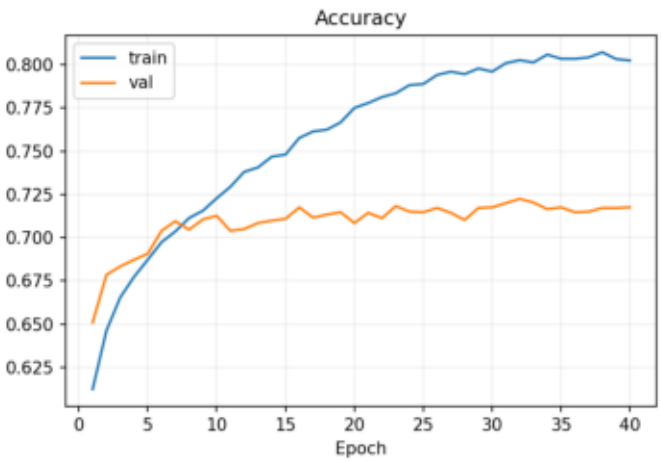
Model	Accuracy	Macro F1	Params	CPU FPS	Target met
cnn	66.41%	0.6415	980,711	495.8	yes
transfer	72.05%	0.7059	4,187,421	11.8	yes

Targets: M1 >= 60%, M2 >= 65%. CPU FPS is batch-one model forward only (float32; M1 1x48x48, M2 3x112x112); it excludes preprocessing, face detection and camera I/O. Realtime throughput must be measured separately on the demonstration laptop. Benchmark CPU: x86_64, threads=4.

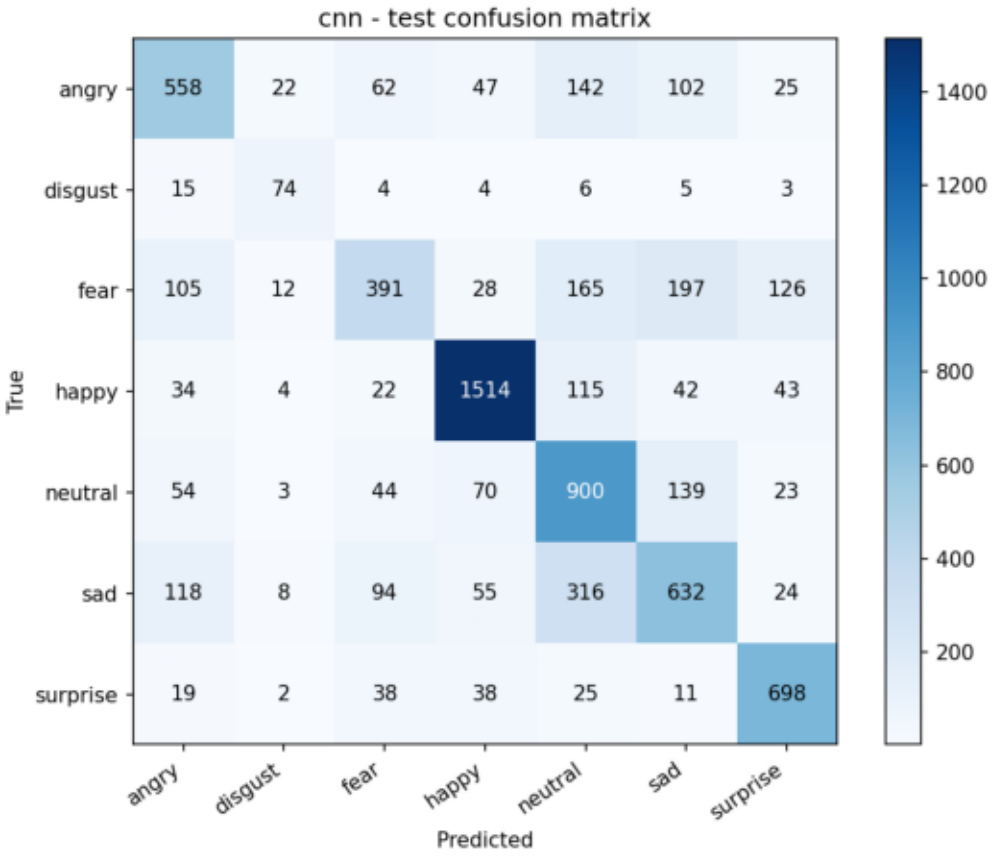


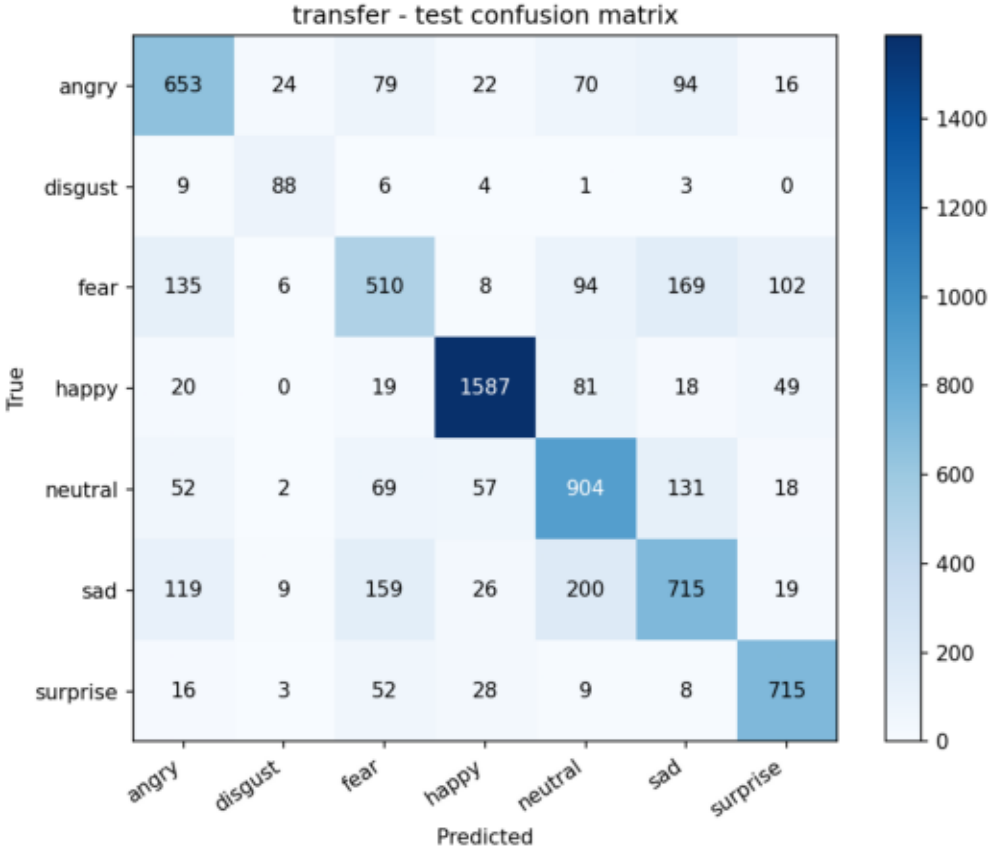






Confusion matrices on held-out test data





CNN - misclassified faces and analysis



1. angry -> sad. Assigned sad 79.1%, versus angry 4.5% (gap 74.6%). Inspect the shared facial cues and whether this label is visually ambiguous.

2. disgust -> surprise. Assigned surprise 61.3%, versus disgust 2.3% (gap 59.0%). Inspect the shared facial cues and whether this label is visually ambiguous.

3. fear -> sad. Assigned sad 60.8%, versus fear 10.1% (gap 50.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

4. happy -> angry. Assigned angry 41.2%, versus happy 15.4% (gap 25.8%). Inspect the shared facial cues and whether this label is visually ambiguous.

5. neutral -> angry. Assigned angry 67.9%, versus neutral 10.5% (gap 57.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

6. sad -> fear. Assigned fear 31.5%, versus sad 22.3% (gap 9.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

7. surprise -> happy. Assigned happy 63.5%, versus surprise 23.7% (gap 39.9%). Inspect the shared facial cues and whether this label is visually ambiguous.

8. angry -> fear. Assigned fear 42.6%, versus angry 33.9% (gap 8.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

9. angry -> neutral. Assigned neutral 33.3%, versus angry 22.2% (gap 11.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

10. angry -> sad. Assigned sad 41.5%, versus angry 26.2% (gap 15.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

Observations above describe score errors and suggest visual checks. Add your own image-specific observations before submission; score patterns alone cannot establish why a face was misclassified.

TRANSFER - misclassified faces and analysis



1. angry -> sad. Assigned sad 87.1%, versus angry 4.7% (gap 82.3%). Confident error; inspect label ambiguity, face crop and domain mismatch.

2. disgust -> fear. Assigned fear 33.5%, versus disgust 18.4% (gap 15.1%). Inspect the shared facial cues and whether this label is visually ambiguous.

3. fear -> angry. Assigned angry 70.4%, versus fear 8.4% (gap 61.9%). Inspect the shared facial cues and whether this label is visually ambiguous.

4. happy -> neutral. Assigned neutral 83.3%, versus happy 14.5% (gap 68.8%). Confident error; inspect label ambiguity, face crop and domain mismatch.

5. neutral -> angry. Assigned angry 79.2%, versus neutral 7.2% (gap 72.1%). Inspect the shared facial cues and whether this label is visually ambiguous.

6. sad -> fear. Assigned fear 46.5%, versus sad 21.0% (gap 25.5%). Inspect the shared facial cues and whether this label is visually ambiguous.

7. surprise -> fear. Assigned fear 70.8%, versus surprise 28.4% (gap 42.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

8. angry -> neutral. Assigned neutral 80.1%, versus angry 7.8% (gap 72.4%). Confident error; inspect label ambiguity, face crop and domain mismatch.

9. angry -> disgust. Assigned disgust 66.9%, versus angry 8.7% (gap 58.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

10. angry -> happy. Assigned happy 49.2%, versus angry 4.4% (gap 44.8%). Inspect the shared facial cues and whether this label is visually ambiguous.

Observations above describe score errors and suggest visual checks. Add your own image-specific observations before submission; score patterns alone cannot establish why a face was misclassified.

Realtime system and demonstration

Pipeline: OpenCV VideoCapture -> Haar frontal-face detection -> crop each face with a small margin -> shared grayscale preprocessing -> batched PyTorch inference -> softmax -> bounding boxes, labels, confidence and FPS.

The command line selects cnn/transfer and accepts a phone URL, webcam, image or file video. Connection failures trigger bounded retries and a clear error. Low-confidence predictions are labelled uncertain. Press s to save the phone-camera demonstration.

The on-screen FPS includes capture and processing, but excludes display and output encoding. It depends on image resolution, number of faces, CPU threads and Wi-Fi. The 10 FPS CPU requirement must be checked on the student's own laptop.

DEMONSTRATION PENDING: no real phone-camera screenshot was supplied. Run the app, press s, then regenerate this report with --screenshot. This report is a draft until the demonstration and visual error analysis are completed.

Limitations: FER labels can be ambiguous; a visible expression does not reliably reveal a person's internal emotional state. Haar detection is strongest on frontal faces. Lighting, pose, occlusion and the FER-to-phone domain shift can reduce accuracy. Softmax scores are model confidence values, not calibrated probabilities.

Sources: kaggle.com/datasets/msambare/fer2013; github.com/SainingZhang/DDAMFN (DDAMFN++ AffectNet-7); pinned source revision and weight SHA-256 are recorded in pretrained_source.json. Checkpoints, split manifests, class counts, run configuration, CSV metrics and error examples accompany the experiment for reproducibility.